

HCP TARGETING & BRAND PERFORMANCE ANALYTICS

Diabetes Portfolio — Case Study

12-Week Upskilling Project | Commercial Analytics

CMS Medicare Part D 2021-2022 | ~227K HCPs | 83M+ Rx Rows

Python | PostgreSQL | Streamlit | openpyxl | sklearn

Executive Summary

This case study documents a 12-week analytics engineering project building a commercial analytics platform for a diabetes drug portfolio. The platform transforms publicly available Medicare claims data into actionable intelligence across five sales personas — enabling field reps, area managers, regional managers, and national leadership to prioritise HCP engagement, identify territory gaps, and model the financial impact of targeting decisions.

Dimension	Detail
Dataset	CMS Medicare Part D 2021-2022 ~227,000 HCPs 83M+ prescription rows
Additional Source	CMS Open Payments — KOL advisory & speaker fees by NPI
Stack	Python 3.11 Streamlit PostgreSQL (Neon serverless) SQLAlchemy
Deployment	Streamlit Community Cloud — live public URL
Personas Served	Sales Rep Area Manager Regional Manager Head of Sales
Key Outputs	HCP priority scoring Territory ROI model Drug trend analysis White space mapping
Duration	12 weeks — structured upskilling exercise

The platform surfaces insights that took hours of manual spreadsheet work in ~3 seconds of SQL query time — and exposes them through a role-personalised UI that requires no SQL knowledge from the end user.

1. Problem Statement

Commercial pharma field teams face a persistent prioritisation problem. With hundreds of HCPs in a territory and limited rep capacity, the question is always: who do I call first, what do I say, and when do I call them back? Without data-driven guidance:

- High-prescribing HCPs get over-called while high-potential new entrants go undiscovered
- Area and regional managers cannot spot territory coverage gaps without running manual reports
- Brand intelligence (which drug classes are growing, where prescribers are switching) is disconnected from field planning
- KOL relationships are managed reactively rather than proactively

This project builds the infrastructure to answer all four of those questions — not through bespoke one-off reports, but through a live, interactive platform that updates with each data refresh.

2. Data Sources & Architecture

2.1 Data Sources

Source	Volume	Key Fields Used
CMS Medicare Part D 2022	83M+ Rx rows → aggregated to HCP level	fills, drug_cost, drug_name, specialty, state
CMS Medicare Part D 2021	Aggregated to HCP level	fills_prior_year — for YoY growth calculation
CMS Open Payments 2022	~12M payment records	NPI, total_payment_usd, opinion_leader_payments
NPPES NPI Registry	~8M provider records	npi, specialty, state, city
CDC T2D Prevalence Data	State-level estimates	State T2D prevalence rate — for white space analysis

2.2 Derived Analytics Fields

The raw claims data is enriched with computed analytics fields during ETL:

Field	Logic	Use
targeting_score	$0.5 * (\text{vol_decile} / 10) + 0.3 * (\text{growth_decile} / 10) + 0.2 * (\text{kol_flag})$	Primary key for HCP prioritisation
segment	CASE: HV if vol+growth decile>=8; Growth if growth decile>=5; else low priority	Segmentation based on volume and growth trends
yoy_growth_pct	$(\text{fills_2022} - \text{fills_2021}) / \text{fills_2021} * 100$	Momentum signal
momentum_score	yoy_growth_pct - simulated_prior (NPI-seeded trend)	Territory direction — rising vs declining
churn_risk	Weighted score: yoy decline, low decile, KOL. At-risk flag	At-risk flag detection
priority_label	P1 if score>=0.72; P2 if >=0.50; else P3	Rep-facing priority triage

2.3 Technology Stack

Layer	Technology	Purpose
Database	Neon serverless PostgreSQL + pgBouncer	Persistent HCP + drug trend data; cold-start tolerant
ORM / Query	SQLAlchemy + pandas	Connection pooling, DataFrame transforms
Application	Python 3.11 + Streamlit 1.35	Multi-persona interactive dashboard
Compute	Streamlit Community Cloud	Zero-infra deployment; live public URL
Modelling	openpyxl + Excel formula engine	Territory ROI calculator (Week 10 deliverable)
Version Control	Git + GitHub	Portfolio-ready public repository

3. What Was Built

3.1 Sales Rep View

The rep view is isolated from all other tabs via `st.stop()` — preventing senior personas from loading the rep-specific query logic. Key features:

- Priority 1/2/3 HCP cards colour-coded by targeting score threshold (0.72 / 0.50)
- Pre-call brief generator: contextual talking points derived from segment, KOL status, growth trajectory, and fill volume
- Call due status with traffic-light colour coding: Overdue / Due Soon / OK — per HCP cadence rule
- One-click 'Mark as Called' simulation updating session state — demonstrates CRM write-back pattern
- Recommended Next Action: rule-based engine (6 action types) surfaced at the individual HCP level

3.2 Area Manager View (4 Tabs)

Tab	Content
My Team	Rep scorecards (5 simulated reps via NPI modulo assignment) with P1 HCP count, calls logged, overdue a
Territory Gaps	Uncovered HCPs table: overdue or dark 30+ days. 3 KPIs: total uncovered, high-value uncovered, estimat
Strategic Accounts	KOL accounts + High Value HCPs with >-3% YoY decline. Expandable pre-call briefs per HCP
Territory Health	4 KPIs: P1 coverage, avg targeting score, churn risk rate, momentum score. Segment donut + at-risk table

3.3 Regional Manager View (4 Tabs)

Tab	Content
My AMs	Area Manager scorecards (1 per state: TX, AZ, NM, OK in Southwest region). Grouped bar chart by segme
Regional Trends	YoY fill change bar chart by state. Segment migration cards: rising HCPs (HV + >10% YoY) vs declining H
KOL Intelligence	KOLs split: engagement gap (advisory payments but declining Rx — competitive threat signal) vs healthy K
White Space	Scatter plot: T2D prevalence vs HCP coverage % by state. Bubble sized by HCP count, coloured by oppor

3.4 Head of Sales View (4 Tabs)

Tab	Content
National Scorecard	4 national KPIs + region performance cards (Red/Amber/Green by coverage) + grouped bar by region
Franchise Accounts	Top 50 HCPs nationally by fills_2022. Trend signal, KOL flag, days since last contact. At-risk alert for YoY
Brand Intelligence	Drug trend analysis from <code>v_drug_trends</code> view: prescriber base and fill volume over time by drug class
Strategic Priorities	Left: HV declining HCPs with estimated revenue at risk (\$). Right: new entrant detection (<code>growth_decile</code> >=8

4. Key Analytical Insights Surfaced

The platform surfaced several commercially relevant patterns from the 2021-2022 Medicare data:

Insight	Signal	Implication
KOL engagement gap	KOLs receiving advisory payments but with YoY decline	Competitor may have made a more compelling advisory offer. Requires follow-up
New entrant detection	growth_decile >= 8, fills_2021 <= 50, fills_2022 > 20	Rapidly growing low-base prescribers — earliest stage to establish relationships
Segment migration	Rising: HV + YoY > 10%. Declining: Leading indicator of next quarter segment reclassification — allows for proactive response	
White space by state	T2D prevalence high but HCP coverage low	Market opportunity not proportional to rep deployment. Signals need for new reps
Cadence miss rate	HCPs where days_dark > CALL_CADENCE_DAYS	Systemic coverage gap detectable at territory level — an Area Manager should be alerted

The most commercially impactful insight class is the KOL engagement gap: a KOL with advisory payments but declining Rx is the strongest competitive threat signal in the dataset. It does not appear in standard CRM reporting — it requires the join of claims data with Open Payments, which this platform does automatically.

5. Territory ROI Calculator (Excel Model)

A standalone five-sheet Excel model built in openpyxl quantifies the commercial impact of targeting decisions. All calculations use Excel formulas referencing Inputs sheet cells — no hardcoded values — so the model recalculates dynamically when assumptions change.

Sheet	Purpose	Key Formulas
Cover	Navigation index with hyperlinks to each sheet	
Inputs	All assumptions in one place (blue = Inputs!C6:G)	Capex_COST=\$180, REV_PER_FILL=\$45, BASE_FILLS=120, cadence = Inputs!C25*(365/Inputs!C19)
Segment ROI	Annual ROI by segment with calls/year (Inputs!C25*(365/Inputs!C19),0) for calls/year; =Inputs!C29*Inputs!C30	
Territory Compare	Side-by-side ROI for 4 territory configurations	Cross-sheet references (green) pulling Inputs values
Scenario Analysis	Best/Base/Worst case with sensitivity for Best and Worst	IF(B2="Best",Inputs!C6*1.2,IF(B2="Worst",Inputs!C6*0.8,Inputs!C6))
Priority Matrix	2x2 ROI vs churn risk heatmap for all segments	Conditional formatting: High ROI + Low Risk = Priority 1 action

Model Metric	Value
Total Excel formulas	102
Formula errors on delivery	0
Colour convention	Blue = inputs Black = formulas Green = cross-sheet links
Verification method	LibreOffice headless recalc.py + cell scan for #REF! / #DIV/0! / #VALUE!

6. What This Would Look Like in Production

The prototype uses CMS public data and a shared serverless database. A production deployment for a real commercial team would differ in three key areas:

Dimension	Prototype (Current)	Production
Data	CMS Medicare Part D (public, lagged Specialty Rx)	Specialty Rx Pharma SHS / IQVIA TRx data — weekly refresh, actual brand
Call logs	Simulated via NPI modulo + datetime seeding	Veeva CRM or Salesforce Health Cloud actuals — real call dates and
Database	Neon serverless PostgreSQL (shared)	Dedicated cloud PostgreSQL / Snowflake with role-based access con
Auth	Streamlit persona selector dropdown	SSO via Okta + Salesforce/Veeva identity — persona locked by user
KOL data	CMS Open Payments (all specialties, all brands)	Brand-specific advisory board membership and speaker bureau data
Refresh cadence	Annual (CMS data cycle)	Weekly (SHS) or daily (CRM events)
ML layer	Rule-based scoring (deterministic)	GBR momentum model + XGBoost churn classifier + multi-label NBA

The architecture is deliberately designed so that swapping data sources requires changing only the ETL layer and database connection string — not the application logic. All persona views, scoring functions, and KPI calculations reference column names that remain constant across CMS and commercial data sources.

7. Skills Demonstrated

Domain	Specific Skills
Data Engineering	ETL pipeline design SQL aggregation (window functions, CTEs, VIEWS) PostgreSQL + SQLAlchemy
Analytics	HCP segmentation logic YoY growth analysis KOL engagement analysis White space analysis
Software Engineering	Multi-persona Streamlit app st.stop() persona isolation Session state management Deterministic
Commercial Acumen	Call cadence strategy by segment Revenue at risk quantification New entrant detection logic KC
Financial Modelling	Territory ROI calculator (5-sheet Excel) Industry colour convention Formula-only model (no hardco
ML Strategy	GBR momentum model spec XGBoost churn classifier Multi-label NBA classifier Survival analy

8. What's Next

- Connect a real commercial data feed (IQVIA / SHS) to replace CMS lagged data
- Train the GBR momentum model once CRM call history (2+ years) is available
- Deploy the XGBoost churn classifier with Salesforce Health Cloud write-back
- Add Veeva CRM call log integration to replace simulated last-call dates
- Implement SSO authentication (Okta / Salesforce Identity) to replace dropdown persona selector
- Build the dynamic cadence engine (Weibull survival analysis) on top of actual interaction data

This document is a portfolio / upskilling exercise artefact. All data used is publicly available from the US Centers for Medicare & Medicaid Services. No proprietary, confidential, or client data has been used.